120 Years of Olympic History: Participation and Medal Trends

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Executive Summary

This report looks at 120 years of Olympic history – from the 1896 Athens Games to the 2016 Rio Games - to analyze participation and medal-winning trends. The analysis shows a major rise in the number of women athletes over time, and a continued dominance of wealthy countries in the medal tallies. Australia has a high Olympic performance relative to its population. The findings highlight the evolving inclusivity of the Olympics and the impact of national resources on sporting success.

Introduction

The Olympic Games is the most popular international sports competition that gathers the best sportsmen from all the world every four years. Since their modern revival in Athens in 1896, the Games have expanded enormously in scale, diversity and global importance. Studying historical trends in Olympic participation and performance can shed light on wider trends in gender equality, national development and sporting investment. This report looks at athlete-level data over 120 years of Olympic history to bring out key trends in participation and medal outcomes.

The dataset, from Kaggle (Griffin 2018), contains over 271,000 observations of athletes, sports, events, and medal results from Summer and Winter Games. The analysis is restricted to the Summer Olympics to ensure comparability of similar events.

By analysing both visual and statistical trends, this report aims to answer key questions about how have participation and medal-winning patterns changed across Olympic Games from 1896 to 2016, with a focus on gender representation and top-performing nations

Research Questions

The analysis is guided by four key questions:

- How has overall athlete participation in the Summer Olympic Games changed over time?
- How has female representation among Olympic athletes changed over time?
- Has the proportion of Olympic medals won by female athletes increased over time?
- Has the distribution of Olympic medals among nations become more or less concentrated over time?

Methodology

Data source:

This data analysis makes use of the "120 Years of Olympic History" dataset found on Kaggle (acknowledeged as The Olympic data on www.sports-reference.com) and has a total of 271,116 observations in 15 different variables which contains information on athletes, their participation in Olympic Games, and the medals they won from 1896 to 2016 (Griffin 2018). Only the Summer Olympic Games were considered in order to ensure consistency in the types of events being compared since there are differences between the Summer and Winter Olympic Games in terms of sports, countries that participate, and numbers of athletes. The Age, Height and Weight variables had missing values which were not necessary for the analysis. Medal values of NA indicate non-medallists and were excluded from medal counts. We used tidyverse (Wickham et al. 2019) knitr (Xie 2025) and kableExtra (rgriffin 2018) for reporting and tables. The dataset includes variables:

- ID: Unique number for each athlete
- Name: Athlete's name
- Sex: M or F
- Age: Integer
- Height: In centimeters
- Weight: In kilograms
- Team: Team name
- NOC: National Olympic Committee 3-letter code
- Games: Year and season
- Year: Integer

- City: Host city
- Sport: Sport
- Event: Event
- Medal: Gold, Silver, Bronze, or NA

Visualisation:

To support the analysis, two key visual outputs were created to summarise Olympic performance patterns. Together, they provide both a graphical and tabular representation of medal distribution, enabling clearer interpretation of national-level Olympic performance patterns.

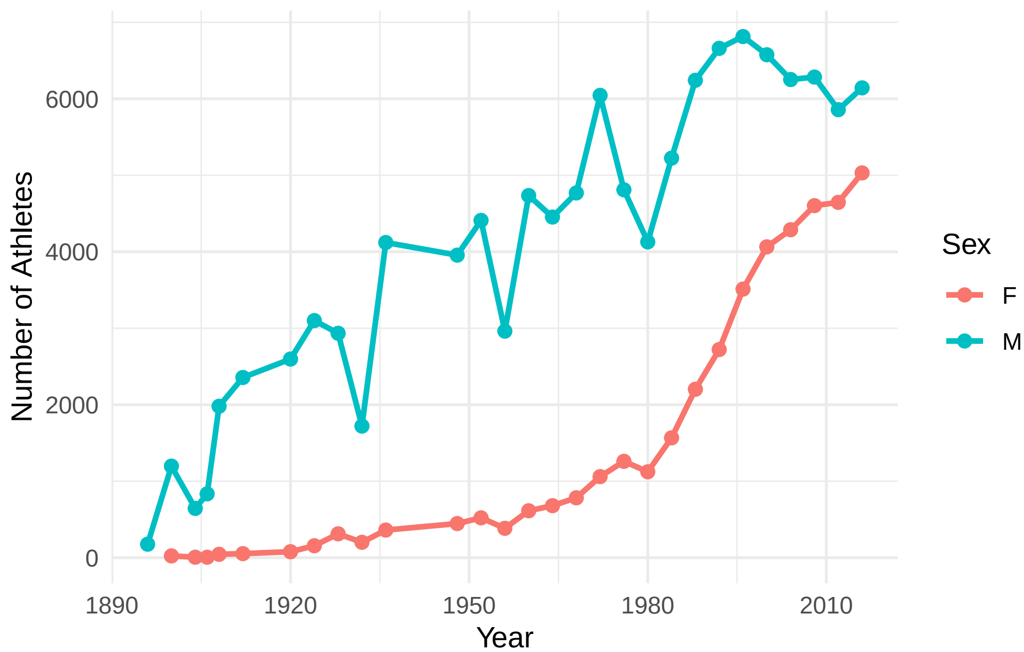


Figure 1: Athlete Participation by Gender in the Summer Olympics (1896–2016)

Figure 1 shows the number of male and female athletes participating in the Summer Olympics from 1896 to 2016. Male participation has consistently exceeded female participation throughout Olympic history. However, female participation has grown substantially since 1900, narrowing the gap between male and female athletes over time.

Table 1: Top 5 Countries by Total Medals (Summer Olympics)

Country	Total Medals
United States	4686
Soviet Union	2061
Germany	1687
Great Britain	1598
France	1408

Table 1 presents the top 5 countries by total medal count in the Summer Olympics. The United States leads with 4,686 medals, followed by Soviet Union with 2,061 medals. Germany, Great Britain and France complete the top 5, all historically powerful Olympic nations.

Statistical Analysis:

Moreover, to analyse and infer how participation and medal-winning patterns have changed across the Olympic Games from 1896 to 2016, we examine four key aspects: overall participation, female representation, female medal success, and medal concentration among nations. Statistical models are used to assess whether these measures exhibit significant trends over time.

1. Overall participation patterns over time: To determine whether Olympic participation has increased over time, a linear regression model is fitted: $Participation_t \sim \beta_0 + \beta_1 *$ Year $+ \epsilon_t$ where $Participation_t$ is the number of athletes participating in each Year. A significant positive coefficient for Year (β_1) would indicate an increase in participation over time.

2. Female Representation over time: To assess whether female representation has changed over time, the proportion of female athletes is analysed using a binomial logistic regression model: $\log\left(\frac{p_t}{1-p_t}\right) \sim \beta_0 + \beta_1 *$ Year $+ \epsilon_t$ where p_t is the proportion of female athletes in each Year. The model is fitted using the yearly counts of female and male athletes. A significantly positive estimate of β_1 would suggest an increasing trend in female representation over time

3. Female Medal Success over time: To investigate whether female athletes have become more successful over time, the proportion of medals won by women is analysed using a binomial logistic regression model: $\log\left(\frac{p_t}{1-p_t}\right) \sim \beta_0 + \beta_1 *$ Year $+ \epsilon_t$ where p_t is the proportion of medals won by female athletes in each Year. The model is fitted using the yearly counts of medals won by female and male athletes. A significantly positive estimate of β_1 would indicate an increasing trend in female medal success over time.

4. Medal Concentration Among Nations: To evaluate whether medal concentration among nations has changed over time, the Herfindahl-Hirschman Index (HHI) is calculated for each Year based on the distribution of medals among countries. A linear regression model

is then fitted: $HHI_t \sim \beta_0 + \beta_1 * \text{Year} + \epsilon_t$ where HHI_t is the Herfindahl-Hirschman Index for each Year. A significant negative coefficient for Year (β_1) would suggest a decrease in medal concentration among nations over time, indicating a more even distribution of medals across countries.

Results

Visualisation

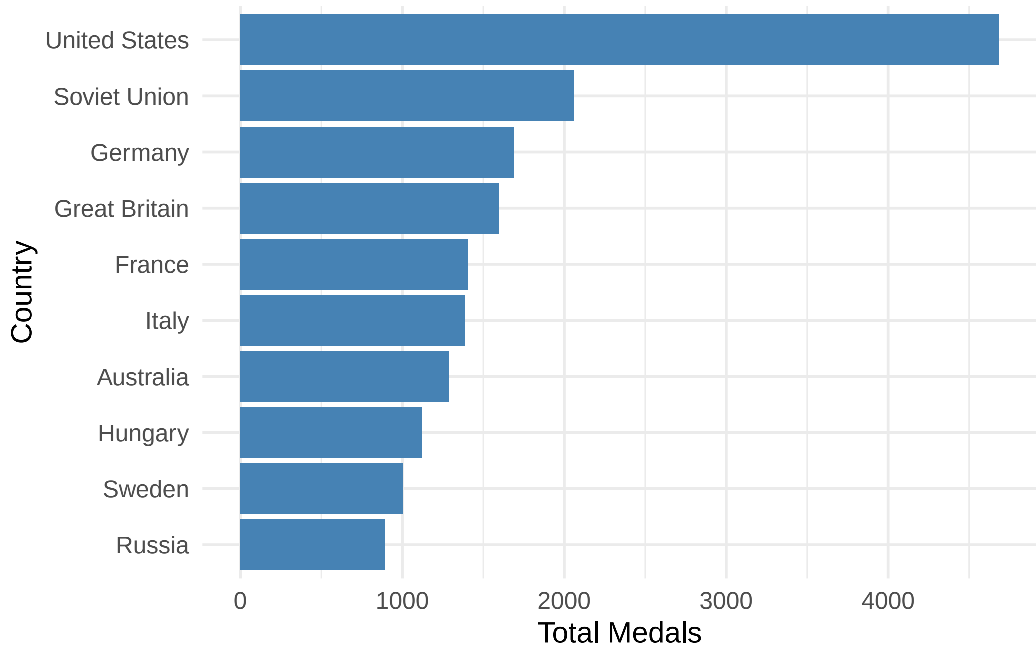


Figure 2: Top 10 Countries by Total Medal Count in the Summer Olympics (1896–2016)

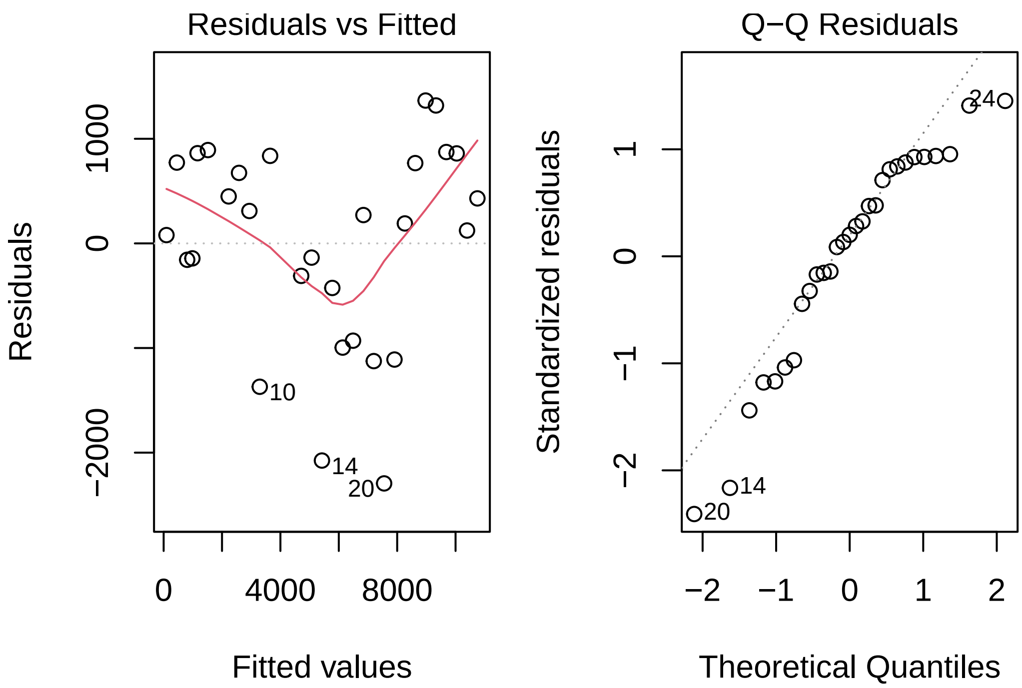
Figure 2 shows the ten countries with the highest total number of medals won through all Summer Olympic Games. Leading is the United States, then comes the Soviet Union and Germany. Australia can be found among the ten most successful countries. There is a noticeable domination of rich countries throughout history, which proves the correlation between national resources and Olympic success.

Statistical Analysis

1. Overall participation patterns over time:

Table 2

Term	Coefficient	p-value	R^2
Intercept	96.6205	0.7825	0.924784415470121
Year	88.7696	0.0000	



(a) Residuals vs Fitted

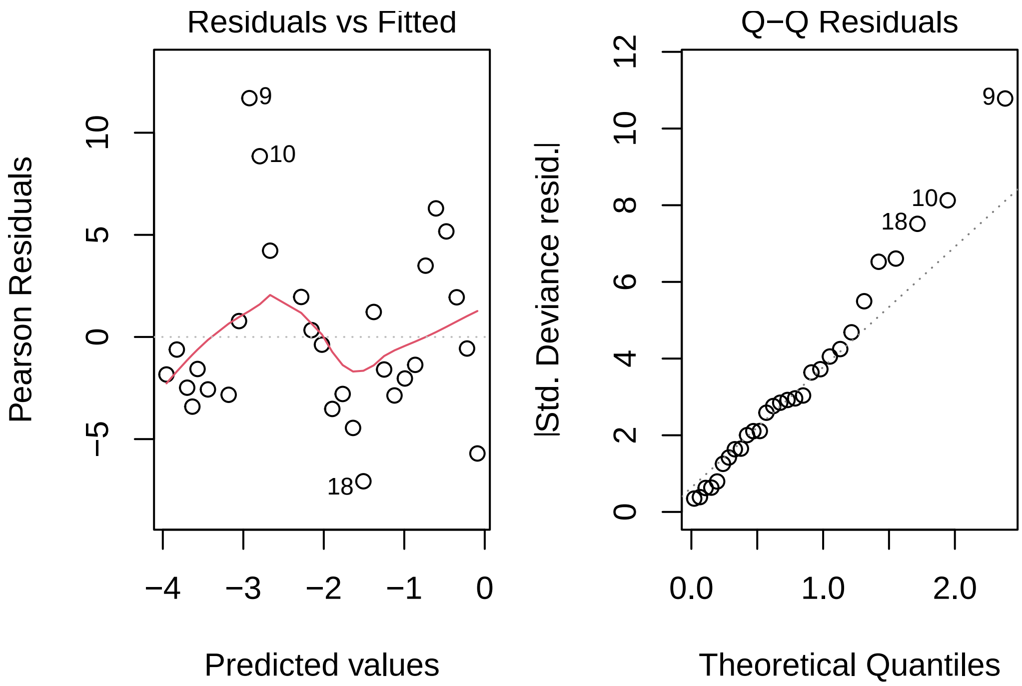
Figure 3: Residual plots for participation model

According to Table 2, the coefficient for Year is positive and statistically significant ($p<0.001$), indicating that Olympic participation has increased over time. The high R-squared value (0.924) suggests that Year explains a large proportion of the variation in athlete participation across Olympic Games. However, the residuals plots show clear patterns rather than random variation indicating that the assumptions of linear regression may not be fully satisfied. This suggests that while there is evidence of a significant upward trend in participation, a more complex model may better capture the relationship over time, such as including polynomial terms or using a non-linear model to account for periods of rapid growth in participation.

2. Female Representation over time:

Table 3

Term	Coefficient	p-value	Pseudo R^2	Null Deviance	Residual Deviance
Intercept	-3.9548	0	0.974581359270954	18562.7511348506	471.83990203946
Year	0.0322	0			



(a) Residuals vs Fitted

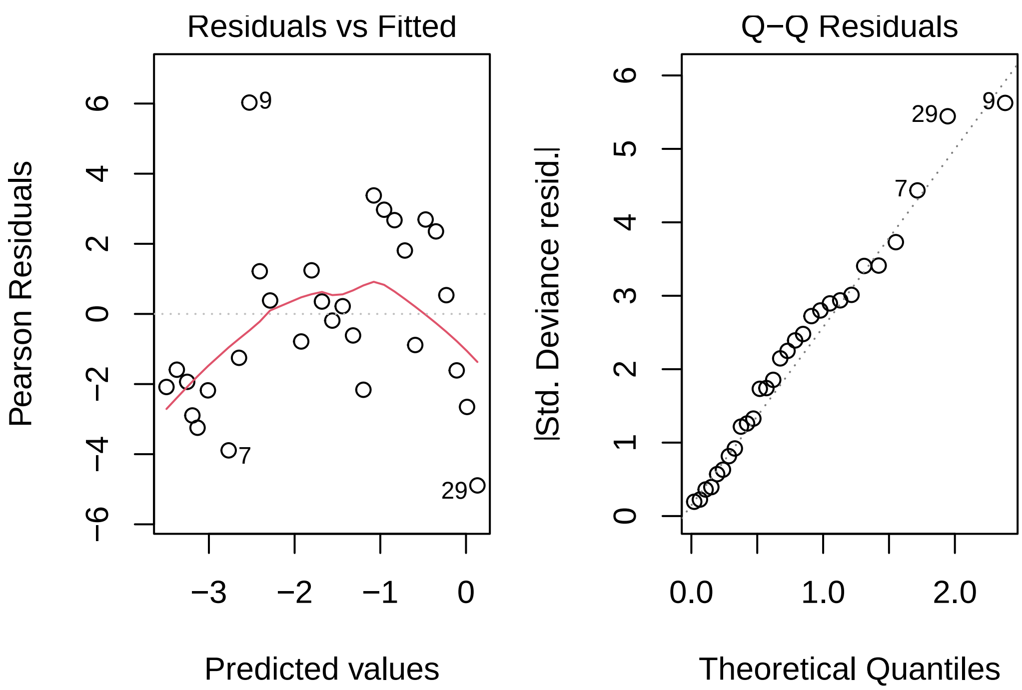
Figure 4: Residual plots for female representation model

The binomial logistic regression indicates that female representation in the Olympic Games has increased slightly over time. Based on Table 3, the coefficient for Year is positive (0.032) and highly significant (p-value < 0.001) suggesting that the proportion of female athletes has increased throughout the study period. The large reduction in deviance from the null model (20058.38) to the fitted model (503.82) indicates that Year explains a substantial amount of the variation in female participation rates. The residual plots do not look reasonably random, indicating that the model may not fully capture the relationship between Year and female representation.

3.

Table 4

Term	Coefficient	p-value	Pseudo R^2	Null Deviance	Residual Deviance
Intercept	-3.4954	0	0.964539393668614	5109.44371104006	181.183972009567
Year	0.0302	0			



(a) Residuals vs Fitted

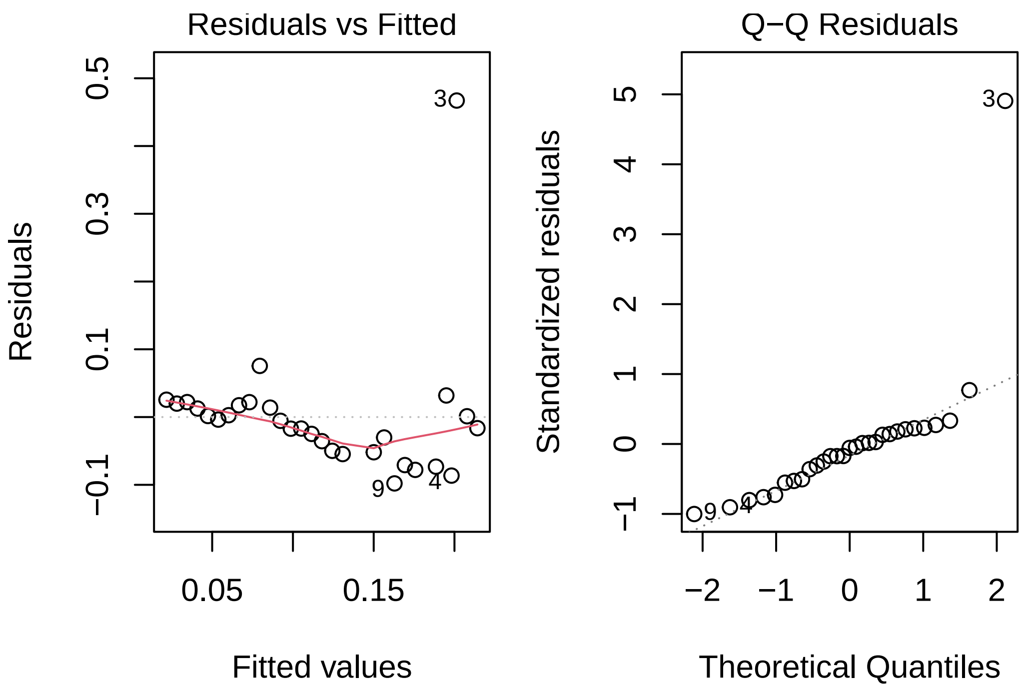
Figure 5: Residual plots for female medal success model

The logistic regression shows an increase in female medal success over time (p-value < 0.001 and $\beta_1 = 0.03$). Table 4 shows the positive coefficient indicating that the proportion of medals won by female athletes has slightly increased throughout Olympic history. Furthermore, the substantial reduction in deviance from the null model (5109.44) to the fitted model (181.18) indicates that Year explains a considerable amount of the variation in female medal success rates. These results provide strong evidence that women's contribution to overall Olympic medal success has increased over time. But the residual plots do not look reasonably random which are the same as the previous two results, indicating that the model may not fully capture the relationship.

4. Medal Concentration Among Nations:

Table 5

Term	Coefficient	p-value	R²
Intercept	0.2142	0.0000	0.274710131339156
Year	-0.0016	0.0035	



(a) Residuals vs Fitted

Figure 6: Residual plots for HHI model

Table 5 indicates a decline in medal concentration over time. The coefficient for Year is negative and statistically significant with $\beta_1 = -0.0016$ and $p = 0.0035$, suggesting that Olympic medals have become less concentrated among a small number of countries and more evenly distributed across participating nations. The model explains approximately 27.4% of the variation in the HHI ($R^2 = 0.274$), indicating that Year is a meaningful predictor of medal concentration, although other factors may also influence the distribution of medals. The residual plots look reasonably random, indicating that the model fits the data well and that the assumptions of linear regression are likely satisfied. This suggests that the observed decline in medal concentration over time is a robust finding, providing evidence of increased competitiveness and diversity in Olympic medal outcomes across nations.

Conclusion and Discussion

The analysis of Summer Olympic Games data from 1896 to 2016 reveals clear and statistically significant changes in participation and medal outcomes over time. Overall athlete participation has increased substantially, reflecting the long-term expansion of the Olympic Games and its growing global reach. Female representation has also increased steadily, with both participation and medal success showing strong upward trends over time. These results provide consistent evidence that the Olympics have become more inclusive, particularly in terms of gender representation.

In addition, the distribution of Olympic medals across nations has become less concentrated over time. The declining Herfindahl–Hirschman Index suggests that medals are no longer dominated by a small group of countries to the same extent as in earlier Olympic Games. This indicates increasing competitiveness and a broader international spread of sporting success, although traditional sporting powers still remain highly successful.

Overall, the findings highlight a clear long-term transformation of the Olympic Games: greater participation, improved gender balance, and more distributed international success. Together, these patterns reflect both the institutional growth of the Olympics and broader global changes in access to sport and investment in athletic development.

Limitations

This study has several limitations. The analysis focuses on long-term trends using Year as the primary explanatory variable and does not account for other factors that may influence participation and medal outcomes, such as population size, economic development, government investment in sport, host-country effects, or geopolitical events. In addition, the linear and logistic regression models may not fully capture non-linear changes in Olympic history, such as major expansions in events or disruptions like world wars. Finally, the HHI measure summarises medal concentration but does not explain the underlying reasons for changes in competitiveness across countries. Future research could explore more complex models that incorporate additional predictors and non-linear effects, as well as qualitative analyses to understand the social and political factors driving changes in Olympic participation and success.

References

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